SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

Trade name	GLUFOSINATE-AMMONIUM SL 150I G
Product code (UVP)	85327623
UN number	UN 2902
Dangerous indication of classifying organizations (if any)	No information
Other national list registration number (if any)	No information

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use	Herbicide

1.3 Details of the supplier of the safety data sheet

Supplier	Bayer Vietnam Ltd,. 118/4 Amata Industrial Park Bien Hoa, Dong Nai Viet Nam
Telefax	+84 251 39 36 951
Responsible Department	Health – Safety – Environment (HSE) +84 251 88 77 120 (ext.:7179/7166) (During business hours only)

1.4 Emergency telephone no.

Vietnam Emergency number	18001775 (During business hours only) +84 251 88 77 120 (ext.:7113)
Global Incident Response Hotline (24h)	+1 (760) 476-3964 (Company 3E for Bayer AG, Crop Science Division)

SECTION 2: HAZARDS IDENTIFICATION PESTICIDES

2.1 Classification of the substance or mixture

Classification in accordance with Vietnamese GHS Regulation on classification, labelling and packaging of substances and mixture, as amended

Acute toxicity: Category 3
H311 Toxic in contact with skin.

Acute toxicity: Category 4
H302 Harmful if swallowed.

Reproductive toxicity: Category 1B
H360Fd May damage fertility. Suspected of damaging the unborn child.

Skin irritation: Category 3
H316 Causes mild skin irritation.

Serious eye damage: Category 1

H318 Causes serious eye damage.

Specific target organ toxicity - repeated exposure: Category 2
H373 May cause damage to organs (Nervous system) through prolonged or repeated exposure if swallowed.

Acute aquatic toxicity: Category 2
H401 Toxic to aquatic life.

Chronic aquatic toxicity: Category 3
H412 Harmful to aquatic life with long lasting effects.

2.2 Label elements

Labelling in accordance with Vietnamese GHS Regulation on classification, labelling and packaging of substances and mixtures, as amendend

Hazard label for supply/use required.

Hazardous components which must be listed on the label:

- Glufosinate ammonium

Signal word: Danger

Hazard statements

H302 Harmful if swallowed.
H311 Toxic in contact with skin.
H316 Causes mild skin irritation.
H318 Causes serious eye damage.
H360Fd May damage fertility. Suspected of damaging the unborn child.
H373 May cause damage to organs (Nervous system) through prolonged or repeated exposure if swallowed.
H412 Harmful to aquatic life with long lasting effects.

Precautionary statements

P201 Obtain special instructions before use.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 Immediately call a POISON CENTER/doctor/ physician.
P391 Collect spillage.
P501 Dispose of contents/container in accordance with local regulation.

2.3 Other hazards

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS PESTICIDES

3.2 Mixtures

Chemical nature

Soluble concentrate (SL)
Glufosinate-ammonium 150 g/l

Hazardous components

Name	CAS-No. / EC-No.	Chemical Formula	Conc. [%]
Glufosinate ammonium	77182-82-2	C5H15N2O4P	13.51
Alkylethersulfate, sodium salt	68891-38-3	No information	> 35 – < 50
1-Methoxy-2-propanol	107-98-2	C4H10O2	> 5 – < 10

SECTION 4: FIRST AID MEASURES

4.1 Description of first aid measures

General advice
Move out of dangerous area. Place and transport victim in stable position (lying sideways). Remove contaminated clothing immediately and dispose of safely. Keep under medical supervision for at least 48 hours.

Inhalation
Move to fresh air. Keep patient warm and at rest. Call a physician or poison control center immediately.

Skin contact
Wash off immediately with soap and plenty of water. Call a physician or poison control center immediately.

Eye contact
Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a physician or poison control center immediately.

Ingestion
Do NOT induce vomiting. Rinse mouth. Call a physician or poison control center immediately.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms
Nausea, Vomiting, Diarrhoea, Abdominal pain, Hypotension, Tremors, Muscular weakness, tachycardia, Convulsions, Unconsciousness, Coma, Respiratory failure

Symptoms may be delayed.

Symptoms and hazards refer to effects observed after intake of significant amounts of the active ingredient(s).

4.3 Indication of any immediate medical attention and special treatment needed

Risks
Watch victim for at least 48 hours because of possible delayed signs of poisoning.

Treatment	Appropriate supportive and symptomatic treatment as indicated by the patient's condition is recommended. In case of ingestion gastric lavage should be considered in cases of significant ingestions only within the first 2 hours. However, the application of activated charcoal and sodium sulphate is always advisable. Forced alkaline diuresis and hemodialysis may be considered. There is no specific antidote. In case of convulsions, a benzodiazepine (e.g. diazepam) should be given according to standard regimens. If not effective, phenobarbital may be used. Contraindication: atropine. Oxygen or artificial respiration if needed. Keep respiratory tract clear. ECG - monitoring (Electrocardiogram). EEG - monitoring (Electroencephalogram). Monitor: respiratory, cardiac, kidney, liver and central nervous system. Keep under medical supervision for at least 48 hours.

SECTION 5: FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable	Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.
Unsuitable	High volume water jet
5.2 Special hazards arising from the substance or mixture	In the event of fire the following may be released:, Hydrogen cyanide (hydrocyanic acid), Carbon monoxide (CO), Carbon dioxide (CO2), Oxides of phosphorus, Nitrogen oxides (NOx)

5.3 Advice for firefighters

The causative agent of fire and explosion (sparks, static electricity, heat, shock, friction, etc.)	(sparks/ heat)
Special protective equipment for firefighters	In the event of fire and/or explosion do not breathe fumes. In the event of fire, wear self-contained breathing apparatus.
Further information	Remove product from areas of fire, or otherwise cool containers with water in order to avoid pressure being built up due to heat. Contain the spread of the fire-fighting media. Do not allow run-off from fire fighting to enter drains or water courses.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Precautions	Avoid contact with spilled product or contaminated surfaces. Use personal protective equipment.
6.2 Environmental precautions	Do not allow to get into surface water, drains and ground water.

6.3 Methods and materials for containment and cleaning up at small and large scale

Methods for cleaning up Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Clean contaminated floors and objects thoroughly, observing environmental regulations.

6.4 Reference to other sections Information regarding safe handling, see section 7.
Information regarding personal protective equipment, see section 8.
Information regarding waste disposal, see section 13.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Advice on safe handling Use only in area provided with appropriate exhaust ventilation.

Advice on protection against fire and explosion Keep away from heat and sources of ignition.

Hygiene measures Avoid contact with skin, eyes and clothing. Keep working clothes separately. Wash hands immediately after work, if necessary take a shower. Remove soiled clothing immediately and clean thoroughly before using again. Garments that cannot be cleaned must be destroyed (burnt).

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers Store in original container. Keep containers tightly closed in a dry, cool and well-ventilated place. Store in a place accessible by authorized persons only. Keep away from direct sunlight. Protect from freezing.

Suitable materials Coex HDPE/EVOH/HDPE
HDPE (high density polyethylene)

7.3 Specific end use(s) Refer to the label and/or leaflet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Components	CAS-No.	Control parameters	Update	Basis
Glufosinate ammonium	77182-82-2	0.9 mg/m3 (TWA)		OES BCS*

*OES BCS: Internal Bayer AG, Crop Science Division "Occupational Exposure Standard"

8.2 Exposure controls

Personal protective equipment
In normal use and handling conditions please refer to the label and/or leaflet. In all other cases the following recommendations would apply.

Respiratory protection Respiratory protection is not required under anticipated circumstances of exposure.
Respiratory protection should only be used to control residual risk of short duration activities, when all reasonably practicable steps have

been taken to reduce exposure at source e.g. containment and/or local extract ventilation. Always follow respirator manufacturer's instructions regarding wearing and maintenance.
Wear respirator with an organic vapours and gas filter mask (protection factor 10) conforming to EN140 type A or equivalent.

Hand protection

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time.
Wash gloves when contaminated. Dispose of when contaminated inside, when perforated or when contamination outside cannot be removed.

Material	Nitrile rubber
Rate of permeability	> 480 min
Glove thickness	> 0.4 mm
Protective index	Class 6
Directive	Protective gloves complying with EN 374.

Eye protection

Wear goggles (conforming to EN166, Field of Use = 5 or equivalent) and faceshield (conforming to EN166, Field of Use = 3 or equivalent).

Skin and body protection

Wear standard coveralls and Category 3 Type 4 suit.
If there is a risk of significant exposure, consider a higher protective type suit.
Wear two layers of clothing wherever possible. Polyester/cotton or cotton overalls should be worn under chemical protection suit and should be professionally laundered frequently.
If chemical protection suit is splashed, sprayed or significantly contaminated, decontaminate as far as possible, then carefully remove and dispose of as advised by manufacturer.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Form	Liquid
Colour	blue to blue green
Odour	weakly pungent
pH	6.8 - 7.8 at 100 % (23 °C)
Melting point/range	No information
Boiling point/boiling range	No information
Flash point	57 °C Does not sustain combustion.
Ignition temperature	No information
Upper explosion limit	No information
Lower explosion limit	No information

Vapour pressure	No information
Evaporation rate	No information
Relative vapour density	No information
Density	ca. 1.11 g/cm³ at 20 °C
Water solubility	at 20 °C miscible
Partition coefficient: n-octanol/water	Glufosinate-ammonium: log Pow: -4.01 at pH 7 Alkylethersulfate, sodium salt: log Pow: 0.3
Surface tension	No information
9.2 Other information	Further safety related physical-chemical data are not known.

SECTION 10: STABILITY AND REACTIVITY PESTICIDES

10.1 Reactivity	
Thermal decomposition	> 200 °C, Heating rate: 10 K/min
10.2 Chemical stability	Stable under normal conditions.
10.3 Possibility of hazardous reactions	No hazardous reactions when stored and handled according to prescribed instructions.
10.4 Conditions to avoid	Extremes of temperature and direct sunlight.
10.5 Incompatible materials	Bases
10.6 Hazardous decomposition products	Ammonia

SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute oral toxicity	LD50 (Rat) 1,730 mg/kg
Acute inhalation toxicity	LC50 (Rat) 2.97 mg/l Exposure time: 4 h Determined in the form of a respirable aerosol. During intended and foreseen applications, no respirable aerosol is formed.
Acute dermal toxicity	LD50 (Rat) 593 mg/kg
Skin corrosion/irritation	Slight irritant effect - does not require labelling. (Rabbit)
Serious eye damage/eye irritation	Severe eye irritation. (Rabbit)
Respiratory or skin sensitisation	Skin: Non-sensitizing. (Guinea pig) OECD Test Guideline 406, Buehler test

Assessment STOT Specific target organ toxicity – single exposure

Glufosinate-ammonium: Based on available data, the classification criteria are not met.
Alkylethersulfate, sodium salt: Based on available data, the classification criteria are not met.

Assessment STOT Specific target organ toxicity – repeated exposure

Glufosinate-ammonium caused neurobehavioral effects and/or neuropathological changes in animal studies. Glufosinate-ammonium was well tolerated in rats and mice but less well tolerated in the dog in subchronic studies.
Alkylethersulfate, sodium salt did not cause specific target organ toxicity in experimental animal studies.

Assessment mutagenicity

Glufosinate-ammonium was not mutagenic or genotoxic in a battery of in vitro and in vivo tests.
Alkylethersulfate, sodium salt was not mutagenic or genotoxic in a battery of in vitro and in vivo tests.

Assessment carcinogenicity

Glufosinate-ammonium was not carcinogenic in lifetime feeding studies in rats and mice.
Alkylethersulfate, sodium salt was not carcinogenic in lifetime feeding studies in rats and mice.

Assessment toxicity to reproduction

Implantation loss occurred in a rat multigeneration study with Glufosinate-ammonium. There were no effects on male fertility.
Alkylethersulfate, sodium salt did not cause reproductive toxicity in a two-generation study in rats.

Assessment developmental toxicity

Glufosinate-ammonium caused developmental toxicity only at dose levels toxic to the dams.
Glufosinate-ammonium caused an increased incidence of post implantation losses.
Alkylethersulfate, sodium salt did not cause developmental toxicity in rats and rabbits.

Aspiration hazard

Based on available data, the classification criteria are not met.

11.2 Other toxic affects

No information

SECTION 12: ECOLOGICAL INFORMATION

12.1 Toxicity

Toxicity to fish	LC50 (Oncorhynchus mykiss (rainbow trout)) 13.4 mg/l Exposure time: 96 h
Toxicity to aquatic invertebrates	EC50 (Daphnia magna (Water flea)) 17.8 mg/l Exposure time: 48 h
Toxicity to aquatic plants	ErC50 (Raphidocelis subcapitata (freshwater green alga)) 8.72 mg/l Growth rate; Exposure time: 72 h
	NOEC (Raphidocelis subcapitata (freshwater green alga)) 1.25 mg/l Exposure time: 72 h

12.2 Persistence and degradability

Biodegradability	Glufosinate-ammonium: Not rapidly biodegradable Alkylethersulfate, sodium salt:

	rapidly biodegradable
Koc	Glufosinate-ammonium: Koc: 2.3 Alkylethersulfate, sodium salt:No data available
Biochemical Oxygen Demand (BOD)	No information
Chemical Oxygen Demand (COD)	No information
Products of biodegradation process	No information
The toxicity level of biodegradable products	No information

12.3 Bioaccumulative potential

Bioaccumulation	Glufosinate-ammonium: Bioconcentration factor (BCF) < 1 Does not bioaccumulate. Alkylethersulfate, sodium salt: Does not bioaccumulate.

12.4 Mobility in soil

Mobility in soil	Glufosinate-ammonium: Highly mobile in soils Alkylethersulfate, sodium salt: soluble in water

12.5 Results of PBT and vPvB assessment

PBT and vPvB assessment	Glufosinate-ammonium: This substance is not considered to be persistent, bioaccumulative and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulative (vPvB). Alkylethersulfate, sodium salt: This substance is not considered to be persistent, bioaccumulative and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulative (vPvB).

12.6 Other adverse effects

Additional ecological information	No further ecological information is available.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Product	In accordance with current regulations and, if necessary, after consultation with the site operator and/or with the responsible authority, the product may be taken to a waste disposal site or incineration plant.
Contaminated packaging	Not completely emptied packagings should be disposed of as hazardous waste.
Waste key for the unused product	14 01 04 agrochemical waste containing dangerous substances
13.2 Information providing destruction (information of law)	Circular 36/2015/BTNMT - management of hazardous wastes

13.3 Product of the process of destruction, treatment measures No information

SECTION 14: TRANSPORT INFORMATION

Regulations on transport of dangerous goods in Vietnam Decree No. 104/2009/ND-CP dated 09/11/2009 for Road transportation of dangerous goods and Decree No. 29/2005/ND-CP dated 10/3/2005 for inland waterways transportation of dangerous goods as amended as per UN MRTDG and IMDG and IATA.

ADR/RID/ADN

14.1 UN number	**2902**
14.2 Proper shipping name	PESTICIDE, LIQUID, TOXIC, N.O.S. (GLUFOSINATE-AMMONIUM SOLUTION)
14.3 Transport hazard class(es)	6.1
14.4 Packaging Group	III
14.5 Environm. Hazardous Mark	NO
Hazard no.	60
Tunnel Code	E

This classification is in principle not valid for carriage by tank vessel on inland waterways. Please refer to the manufacturer for further information.

IMDG

14.1 UN number	**2902**
14.2 Proper shipping name	PESTICIDE, LIQUID, TOXIC, N.O.S. (GLUFOSINATE-AMMONIUM SOLUTION)
14.3 Transport hazard class(es)	6.1
14.4 Packaging Group	III
14.5 Marine pollutant	NO
Segregation group according to 5.4.1.5.11.1	IMDG SEGREGATION GROUP 2 - AMMONIUM COMPOUNDS

IATA

14.1 UN number	**2902**
14.2 Proper shipping name	PESTICIDE, LIQUID, TOXIC, N.O.S. (GLUFOSINATE-AMMONIUM SOLUTION)
14.3 Transport hazard class(es)	6.1
14.4 Packaging Group	III
14.5 Environm. Hazardous Mark	NO

14.6 Special precautions for user and transporter
See sections 6 to 8 of this Safety Data Sheet.

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code
No transport in bulk according to the IBC Code.

SECTION 15: REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Further information

WHO-classification: II (Moderately hazardous)

15.2 Dangerous classification by countries declared, registered

No information

15.3 Status of declaration, registration in the regional countries in the world (list the countries conducted declaration, status of declaration)

No information

SECTION 16: OTHER INFORMATION

Abbreviations and acronyms

ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE	Acute toxicity estimate
CAS-Nr.	Chemical Abstracts Service number
Conc.	Concentration
EC-No.	European community number
ECx	Effective concentration to x %
EINECS	European inventory of existing commercial substances
ELINCS	European list of notified chemical substances
EN	European Standard
EU	European Union
IATA	International Air Transport Association
IBC	International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (IBC Code)
ICx	Inhibition concentration to x %
IMDG	International Maritime Dangerous Goods
LCx	Lethal concentration to x %
LDx	Lethal dose to x %
LOEC/LOEL	Lowest observed effect concentration/level
MARD	Ministry of Agriculture and Rural Development.
MARPOL	MARPOL: International Convention for the prevention of marine pollution from ships
N.O.S.	Not otherwise specified
NOEC/NOEL	No observed effect concentration/level
OECD	Organization for Economic Co-operation and Development
RID	Regulations concerning the International Carriage of Dangerous Goods by Rail
TWA	Time weighted average
UN	United Nations
WHO	World health organisation

The information in the safety data sheet of chemical and Pesticides is compiled based on valid and latest knowledge and on dangerous Pesticides and chemical and should be used to implement measures to prevent the risk, accidents. This data sheet complements the user's instructions but does

not replace them.

Dangerous Pesticides in this slip may have other hazardous properties depending on the circumstances of use and exposure.

Changes since the last version are highlighted in the margin. This version replaces all previous versions.